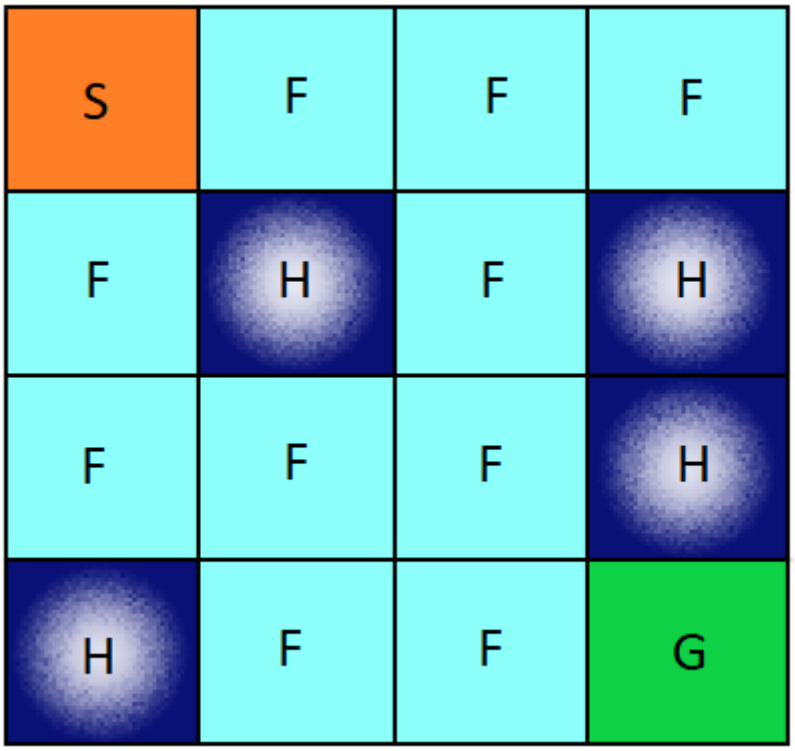
Frozenlake Case

The Frozen Lake environment is a 4×4 grid which contain four possible areas: Safe (S), Frozen (F), Hole (H) and Goal (G). The agent moves around the grid until it reaches the goal or the hole. If it falls into the hole, it has to start from the beginning and is rewarded the value 0. The process continues until it learns from every mistake and reaches the goal eventually. Here is visual description of the Frozen Lake grid (4×4):



S: Safe
F: Frozen
H: Hole
G: Goal

The agent in the environment has four possible moves — Up, Down, Left and Right. We will be implementing one of the Reinforcement Learning techniques, Q-Learning, here. This environment will allow the agent to move accordingly. There could be a random action happening once every a few episodes — let’s say the agent is slipping in different directions because it is hard to walk on a frozen surface. Considering this situation, we need to allow some random movement at first, but eventually try to reduce its probability. This way we can correct the error caused by minimising the loss.

This grid has 16 possible blocks where the agent will be at a given time. At the current state, the agent will have four possibilities of deciding the next state. From the current state, the agent can move in four directions as mentioned which gives us a total of 16×4 possibilities. These are our weights and we will update them according to the movement of our agent.

The goal of this game is to go from the starting state (S) to the goal state (G) by walking only on frozen tiles (F) and avoid holes (H). However, the ice is slippery, so you won't always move in the direction you intend (stochastic environment).